

SolarTwin AI

Enterprise Solar Assessment Report

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Project Information

Parameter	Value
Location	Airoli, Airoli Station Road, Airoli Sector 2, Airoli, Navi Mumbai, Thane
Roof Area	214.52 m²
Panels	56
Capacity	30.80 kW
Orientation	Portrait

Weather

Parameter	Value
Temperature	28.7 °C
Cloud Cover	100 %
Wind	14.3 km/h
Humidity	81 %

Solar Production

Parameter	Value
Annual Energy	45885 kWh
Daily Energy	125.7 kWh
Annual Saving	■ 367,082
Payback	5.0 Years
25 Year Profit	■ 7,329,042

AI Recommendations

- Large rooftop detected. Industrial-scale installation is feasible.
- Portrait layout provides the best packing efficiency.
- Additional roof optimization may increase installed capacity.
- Frequent cloud cover may reduce annual generation.
- Large-capacity solar installation identified.
- Clean solar panels every 2–3 months for optimal efficiency.

- Perform annual electrical and structural inspections.

Executive Conclusion

SolarTwin AI concludes that the analysed rooftop is suitable for a photovoltaic installation of approximately 30.80 kW. The estimated annual production is 45885 kWh, with estimated annual savings of ■ 367,082. The generated layout maximizes the usable roof area while considering rooftop geometry and detected obstacles. The proposed design is recommended for further engineering validation before installation.